

Assignment: Duality, Convexity, and Optimization Foundations

Complete Solutions with Detailed Explanations

Contents

1	Short-Answer Questions (Q1–Q12)	2
2	Long-Answer Questions (Q13–Q18)	7

1 Short-Answer Questions (Q1–Q12)

Q1 (Short Answer) — Standard Form of an Optimization Problem

Write the *standard form* of an optimization problem. Clearly label: (i) the optimization variable, (ii) the objective function, (iii) inequality constraints, and (iv) equality constraints.

Solution: The **standard form** of a (minimization) optimization problem is:

$$\begin{aligned} & \underset{x}{\text{minimize}} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & && h_j(x) = 0, \quad j = 1, \dots, p. \end{aligned}$$

- (i) **Optimization variable:** $x \in \mathbb{R}^d$ — the quantity we are trying to find.
- (ii) **Objective function:** $f_0 : \mathbb{R}^d \rightarrow \mathbb{R}$ — the quantity we are minimizing.
- (iii) **Inequality constraints:** $f_i(x) \leq 0$ for $i = 1, \dots, m$. Any “ $\geq$ ” constraint $g_i(x) \geq 0$ is rewritten as $-g_i(x) \leq 0$.
- (iv) **Equality constraints:** $h_j(x) = 0$ for $j = 1, \dots, p$. These must be *affine* ($h_j(x) = a_j^\top x - b_j$) for the problem to be convex.

Why this form? It is the universal template: every bound, box constraint, or linear constraint can be cast into this shape. The inequality convention “ ≤ 0 ” (rather than “ $\leq c$ ”) is a matter of standardization — just move c to the left.

Q2 (Short Answer) — Domain and Feasible Set

Define the *domain* of an optimization problem in standard form using the domains of the objective/constraint functions. Then define the *feasible set*. What does it mean for the problem to be *feasible*?

Solution: **Domain** of the problem:

$$\mathcal{D} = \text{dom}(f_0) \cap \bigcap_{i=1}^m \text{dom}(f_i) \cap \bigcap_{j=1}^p \text{dom}(h_j).$$

This is the set of all points x where every function appearing in the problem is defined (the implicit constraint). Note: a point can be in $\mathcal{D}$ but still violate constraints.

Feasible set (also called the constraint set):

$$\mathcal{F} = \{x \in \mathcal{D} \mid f_i(x) \leq 0 \ \forall i, \ h_j(x) = 0 \ \forall j\}.$$

This is the set of all x that are both in the domain *and* satisfy all constraints.

Feasibility: The problem is *feasible* if $\mathcal{F} \neq \emptyset$, i.e., at least one point satisfies every constraint. If $\mathcal{F} = \emptyset$, the problem is *infeasible* and there is no solution.

Q3 (Short Answer) — Optimal Value, Optimal Point, Attained

Define:

- (a) the optimal value p^* ,
- (b) an optimal point,
- (c) what it means for the optimal value to be *attained*.

Solution:

- (a) **Optimal value:**

$$p^* = \inf_{x \in \mathcal{F}} f_0(x).$$

It is the greatest lower bound of f_0 over the feasible set. Conventions: $p^* = +\infty$ if $\mathcal{F} = \emptyset$ (infeasible); $p^* = -\infty$ if f_0 is unbounded below on $\mathcal{F}$ (unbounded).

- (b) **Optimal point (solution):** A point $x^* \in \mathcal{F}$ such that $f_0(x^*) = p^*$. In other words, x^* is feasible and achieves the infimum.
- (c) **Attained:** The optimal value is *attained* if there exists at least one optimal point x^* , i.e., the infimum is a minimum. The infimum may exist (be finite) without being attained — for example, $\min_{x>0} \frac{1}{x}$ has $p^* = 0$ but no $x > 0$ achieves it.

Q4 (Short Answer) — ε -Suboptimal, Active/Inactive/Redundant Constraints

In the context of constrained optimization, define:

- (a) an ε -suboptimal point,
- (b) an *active* inequality constraint at a feasible point,
- (c) an *inactive* inequality constraint at a feasible point,
- (d) a *redundant* constraint.

Solution:

- (a) **ε -suboptimal point:** A feasible point $\tilde{x} \in \mathcal{F}$ such that

$$f_0(\tilde{x}) \leq p^* + \varepsilon.$$

It is within ε of optimal. Useful in practice because algorithms cannot usually reach p^* exactly.

- (b) **Active constraint:** Inequality constraint $f_i(x) \leq 0$ is *active* at feasible $\bar{x}$ if $f_i(\bar{x}) = 0$. The constraint is *tight*; the point sits exactly on the constraint boundary.
- (c) **Inactive constraint:** f_i is *inactive* at $\bar{x}$ if $f_i(\bar{x}) < 0$. There is slack — the constraint is not binding, and removing it locally would not change the feasible set near $\bar{x}$.
- (d) **Redundant constraint:** A constraint is *redundant* if removing it does not change the feasible set $\mathcal{F}$. Equivalently, every feasible point already satisfies this constraint strictly (or it is implied by other constraints). Redundant constraints add no information but can slow down solvers.

Q5 (Short Answer) — Converting Maximization to Minimization

Explain how to convert a *maximization* problem into an equivalent *minimization* problem without changing the feasible set.

Solution: To maximize $f_0(x)$ subject to constraints, simply negate the objective:

$$\max_{x \in \mathcal{F}} f_0(x) \iff \min_{x \in \mathcal{F}} [-f_0(x)].$$

Why this works:

$$\max_{x \in \mathcal{F}} f_0(x) = -\min_{x \in \mathcal{F}} (-f_0(x)).$$

The feasible set $\mathcal{F}$ is unchanged (only the objective changes). The optimal point x^* is identical for both problems; only the sign of the optimal value flips: $p_{\max}^* = -p_{\min}^*$.

Important: If f_0 was convex, $-f_0$ is concave, so the minimization of a concave function is in general not a convex problem. To keep convexity for maximization, one maximizes a *concave* function (equivalently, minimizes a convex function).

Q6 (Short Answer) — Box Constraints in Standard Form

Consider box constraints $l_i \leq x_i \leq u_i$ for $i = 1, \dots, n$. Rewrite these constraints in standard form $f_i(x) \leq 0$.

Solution: Each two-sided bound $l_i \leq x_i \leq u_i$ gives *two* standard-form inequality constraints:

$$\begin{aligned} x_i \leq u_i &\implies f_i^{(+)}(x) = x_i - u_i \leq 0, \\ x_i \geq l_i &\implies f_i^{(-)}(x) = l_i - x_i \leq 0. \end{aligned}$$

For all $i = 1, \dots, n$ we get $2n$ constraints in total:

$$x_i - u_i \leq 0 \quad \text{and} \quad l_i - x_i \leq 0, \quad i = 1, \dots, n.$$

Each $f_i^{(\pm)}$ is a linear (hence convex) function of x , so box constraints are convex constraints compatible with a convex program.

Q7 (Short Answer) — Slack Variable Transformation

Let an optimization problem have inequality constraints $f_i(x) \leq 0$. Explain the *slack-variable* transformation that replaces each inequality with an equality plus a nonnegativity constraint.

Solution: For each inequality $f_i(x) \leq 0$, introduce a **slack variable** $s_i \geq 0$ and write:

$$f_i(x) + s_i = 0, \quad s_i \geq 0.$$

This is equivalent: if $f_i(x) \leq 0$ then $s_i = -f_i(x) \geq 0$ satisfies the equality; conversely, if $f_i(x) + s_i = 0$ with $s_i \geq 0$ then $f_i(x) = -s_i \leq 0$.

Result: The problem becomes

$$\min_x f_0(x) \quad \text{s.t.} \quad f_i(x) + s_i = 0, \quad s_i \geq 0 \quad (i = 1, \dots, m), \quad h_j(x) = 0 \quad (j = 1, \dots, p).$$

The new variable vector is $(x, s_1, \dots, s_m)$, with m equality constraints and m nonnegativity constraints $s_i \geq 0$. This is useful in linear programming (where f_i are affine) and for interior-point methods.

Q8 (Short Answer) — Convex Set Definition and Intersection Property

State the definition of a *convex set* $C \subseteq \mathbb{R}^d$. Then state (without proof) the intersection property: if C_1 and C_2 are convex, what can you say about $C_1 \cap C_2$?

Solution: Definition: A set $C \subseteq \mathbb{R}^d$ is *convex* if for any two points $x, y \in C$ and any $\theta \in [0, 1]$, the convex combination also lies in C :

$$\theta x + (1 - \theta)y \in C \quad \forall x, y \in C, \forall \theta \in [0, 1].$$

Geometrically: the line segment joining any two points in C lies entirely within C .

Intersection property: The intersection of any collection of convex sets is convex. In particular, if C_1 and C_2 are convex, then $C_1 \cap C_2$ is convex.

Why this matters: Feasible sets of convex programs are intersections of convex sets (each constraint $\{x : f_i(x) \leq 0\}$ is convex when f_i is convex), so the feasible set is convex — which is what makes convex programs tractable.

Q9 (Short Answer) — Convex Function Definition and First-Order Characterization

State the definition of a *convex function* $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$. Also state one common *first-order* characterization of convexity when f is differentiable.

Solution: Definition (Jensen's inequality form): f is convex if $\text{dom}(f)$ is a convex set and for all $x, y \in \text{dom}(f)$ and $\theta \in [0, 1]$:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

The function value at a convex combination is at most the corresponding convex combination of function values. The graph lies *below* any chord.

First-order characterization (gradient inequality): If f is differentiable on its (open convex) domain, then f is convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \text{dom}(f).$$

The tangent hyperplane at any point is a *global underestimator* of f . Equivalently, the graph of f lies above every tangent plane.

Second-order characterization (for twice-differentiable f): f is convex $\iff \nabla^2 f(x) \succeq 0$ (positive semidefinite Hessian) for all $x \in \text{dom}(f)$.

Q10 (Short Answer) — Indicator Function

Define the *indicator function* $\mathbf{1}_X$ of a set X (as used in convex analysis/optimization). How can $\mathbf{1}_X$ be used to rewrite a constrained problem $\min_{x \in X} f(x)$ as an unconstrained problem?

Solution: Indicator function:

$$\mathbf{1}_X(x) = \begin{cases} 0 & \text{if } x \in X, \\ +\infty & \text{if } x \notin X. \end{cases}$$

It assigns zero cost inside the set and infinite cost outside, effectively “forbidding” points outside X .

Unconstrained reformulation:

$$\min_{x \in X} f(x) = \min_{x \in \mathbb{R}^d} [f(x) + \mathbf{1}_X(x)].$$

The right-hand side has no constraints — the feasibility is enforced implicitly by $\mathbf{1}_X$. Any point with $x \notin X$ gives objective value $+\infty$ and is thus never optimal.

Convexity: $\mathbf{1}_X$ is a convex function if and only if X is a convex set (proved in Q13). Therefore, the reformulation is a convex unconstrained problem whenever f is convex and X is convex.

Q11 (Short Answer) — Norm Axioms and Dual Norm

Define a *norm* $\|\cdot\|$ on $\mathbb{R}^d$ via its three standard axioms. Then define the *dual norm* $\|\cdot\|_*$.

Solution: Norm axioms: A function $\|\cdot\| : \mathbb{R}^d \rightarrow \mathbb{R}$ is a norm if:

- (i) **Positive definiteness:** $\|x\| \geq 0$ for all x , with $\|x\| = 0 \Leftrightarrow x = 0$.
- (ii) **Absolute homogeneity:** $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{R}$, $x \in \mathbb{R}^d$.
- (iii) **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^d$.

Dual norm: Given a norm $\|\cdot\|$, its dual norm is

$$\|y\|_* = \sup\{y^\top x \mid \|x\| \leq 1\} = \sup_{x \neq 0} \frac{y^\top x}{\|x\|}.$$

It measures the maximum inner product y can achieve with unit-norm vectors.

Examples:

- The dual of ℓ_p is ℓ_q where $\frac{1}{p} + \frac{1}{q} = 1$ (Hölder conjugates).
- The dual of ℓ_2 is ℓ_2 (self-dual): $\|y\|_* = \|y\|_2$.
- The dual of ℓ_1 is ℓ_∞ : $\|y\|_* = \|y\|_\infty$.
- The dual of ℓ_∞ is ℓ_1 .

The dual norm also satisfies the generalized Cauchy-Schwarz: $y^\top x \leq \|y\|_* \|x\|$.

Q12 (Short Answer) — Lagrangian and Dual Function

Give the Lagrangian for the standard-form problem

$$\min_x f_0(x) \quad \text{s.t.} \quad f_i(x) \leq 0 \quad (i = 1, \dots, m), \quad h_j(x) = 0 \quad (j = 1, \dots, p).$$

Define the dual function $g(\lambda, \nu)$. What sign restriction is required on λ ?

Solution: Lagrangian: Introduce multipliers $\lambda_i \geq 0$ for each inequality and $\nu_j \in \mathbb{R}$ for each equality:

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{j=1}^p \nu_j h_j(x).$$

The Lagrangian relaxes the constraints into the objective, penalizing violations via the multipliers.

Dual function: Minimize the Lagrangian over x :

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \inf_{x \in \mathcal{D}} \left[f_0(x) + \sum_i \lambda_i f_i(x) + \sum_j \nu_j h_j(x) \right].$$

g is always *concave* in (λ, ν) (as the pointwise infimum of affine functions of (λ, ν)), even if the original problem is nonconvex.

Sign restriction on λ : We require $\lambda \geq 0$ (componentwise), i.e., $\lambda_i \geq 0$ for all i . This ensures $g(\lambda, \nu) \leq p^*$ (weak duality, proved next). If $\lambda_i < 0$, the Lagrangian could be decreased by making $f_i(x)$ large and positive, breaking the lower-bound property. No sign restriction is needed on ν (equality constraints can be violated in either direction).

2 Long-Answer Questions (Q13–Q18)

Q13 (Long Answer) — Indicator Function is Convex iff X is Convex

Prove the following claim: the indicator function $\mathbf{1}_X$ is convex if and only if X is a convex set. Your proof should go in both directions.

Solution: We prove both directions.

($\Rightarrow$) **Assume $\mathbf{1}_X$ is convex; prove X is convex.**

Take any $x, y \in X$ and $\theta \in [0, 1]$. Since $x, y \in X$:

$$\mathbf{1}_X(x) = 0, \quad \mathbf{1}_X(y) = 0.$$

By convexity of $\mathbf{1}_X$:

$$\mathbf{1}_X(\theta x + (1 - \theta)y) \leq \theta \mathbf{1}_X(x) + (1 - \theta) \mathbf{1}_X(y) = \theta \cdot 0 + (1 - \theta) \cdot 0 = 0.$$

Since $\mathbf{1}_X$ only takes values 0 or $+\infty$, the inequality $\mathbf{1}_X(\theta x + (1 - \theta)y) \leq 0$ forces

$$\mathbf{1}_X(\theta x + (1 - \theta)y) = 0,$$

which means $\theta x + (1 - \theta)y \in X$. Since $x, y \in X$ and $\theta \in [0, 1]$ were arbitrary, X is convex. ✓

($\Leftarrow$) **Assume X is convex; prove $\mathbf{1}_X$ is convex.**

We need to verify: for all $x, y \in \mathbb{R}^d$ (the ambient space, which is the domain of $\mathbf{1}_X$ extended to $+\infty$ outside X) and $\theta \in [0, 1]$:

$$\mathbf{1}_X(\theta x + (1 - \theta)y) \leq \theta \mathbf{1}_X(x) + (1 - \theta) \mathbf{1}_X(y).$$

Case 1: $x \notin X$ or $y \notin X$. Then $\mathbf{1}_X(x) = +\infty$ or $\mathbf{1}_X(y) = +\infty$, so the right side is $+\infty$. The left side is at most $+\infty$. The inequality holds trivially.

Case 2: $x \in X$ and $y \in X$. Then $\mathbf{1}_X(x) = \mathbf{1}_X(y) = 0$, so the right side is 0. Since X is convex and $x, y \in X$, we have $\theta x + (1 - \theta)y \in X$, so the left side is also 0. The inequality $0 \leq 0$ holds. ✓

In both cases the convexity inequality holds, so $\mathbf{1}_X$ is convex. ■

Key insight: The indicator function encodes set membership as a convexity condition. Convex analysis uses this bridge constantly: constrained problems become unconstrained ones, but the indicator carries the geometry of the feasible set.

Q14 (Long Answer) — Pointwise Maximum of Convex Functions is Convex

Let $\{f(\cdot; y)\}_{y \in \mathcal{Y}}$ be a family of convex functions in x (for each fixed y). Define

$$F(x) := \sup_{y \in \mathcal{Y}} f(x; y).$$

Prove that F is convex (assuming $F(x) > -\infty$ for all x in its domain).

Solution: **Claim:** $F(x) = \sup_{y \in \mathcal{Y}} f(x; y)$ is convex.

Proof.

Let $x_1, x_2 \in \text{dom}(F)$ (so $F(x_1) < +\infty$ and $F(x_2) < +\infty$), and let $\theta \in [0, 1]$.

For each fixed $y \in \mathcal{Y}$, $f(\cdot; y)$ is convex, so:

$$f(\theta x_1 + (1 - \theta)x_2; y) \leq \theta f(x_1; y) + (1 - \theta)f(x_2; y).$$

Taking the supremum over all $y \in \mathcal{Y}$ on both sides, and using the fact that the supremum of a sum is at most the sum of suprema:

$$F(\theta x_1 + (1 - \theta)x_2) = \sup_{y \in \mathcal{Y}} f(\theta x_1 + (1 - \theta)x_2; y) \leq \sup_{y \in \mathcal{Y}} [\theta f(x_1; y) + (1 - \theta)f(x_2; y)].$$

Now we use the bound $\theta f(x_1; y) + (1 - \theta)f(x_2; y) \leq \theta \sup_y f(x_1; y) + (1 - \theta) \sup_y f(x_2; y) = \theta F(x_1) + (1 - \theta)F(x_2)$ for every fixed y (since both terms are bounded above by their respective sups). Therefore:

$$\sup_{y \in \mathcal{Y}} [\theta f(x_1; y) + (1 - \theta)f(x_2; y)] \leq \theta F(x_1) + (1 - \theta)F(x_2).$$

Combining:

$$F(\theta x_1 + (1 - \theta)x_2) \leq \theta F(x_1) + (1 - \theta)F(x_2).$$

This is exactly the definition of convexity, so F is convex. ■

Important consequences:

- The *maximum* of finitely many convex functions is convex (take $\mathcal{Y}$ finite).
- The *support function* $h_C(y) = \sup_{x \in C} y^\top x$ is convex in y (each $y^\top x$ is linear/convex in y).
- The *conjugate function* $f^*(y) = \sup_x (y^\top x - f(x))$ is always convex in y .
- *Norms* can be written as $\|x\| = \sup_{\|y\|_* \leq 1} y^\top x$, hence are convex.

Q15 (Long Answer) — Affine Composition Preserves Convexity

Let $f : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and define $g : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ by

$$g(x) = f(Ax + b),$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ are fixed. Prove that g is convex. Then apply your result to argue that $x \mapsto \frac{1}{2} \|Ax - b\|_2^2$ is convex.

Solution: **Proof that g is convex.**

Let $x_1, x_2 \in \text{dom}(g)$ and $\theta \in [0, 1]$. Then $Ax_1 + b, Ax_2 + b \in \text{dom}(f)$.

Compute the argument at the convex combination:

$$A(\theta x_1 + (1 - \theta)x_2) + b = \theta(Ax_1 + b) + (1 - \theta)(Ax_2 + b).$$

(This is the key: the affine map sends a convex combination to a convex combination.)

Since f is convex and $Ax_1 + b, Ax_2 + b \in \text{dom}(f)$:

$$\begin{aligned} g(\theta x_1 + (1 - \theta)x_2) &= f(A(\theta x_1 + (1 - \theta)x_2) + b) \\ &= f(\theta(Ax_1 + b) + (1 - \theta)(Ax_2 + b)) \\ &\leq \theta f(Ax_1 + b) + (1 - \theta)f(Ax_2 + b) \\ &= \theta g(x_1) + (1 - \theta)g(x_2). \end{aligned}$$

Hence g is convex. ■

Application: $\phi(x) = \frac{1}{2}\|Ax - b\|_2^2$.

Step 1: The squared ℓ_2 norm $f(u) = \frac{1}{2}\|u\|_2^2 = \frac{1}{2}u^\top u$ is convex. Proof: $\nabla^2 f(u) = I \succeq 0$, or directly:

$$f(\theta u + (1 - \theta)v) = \frac{1}{2}\|\theta u + (1 - \theta)v\|^2 \leq \theta \frac{1}{2}\|u\|^2 + (1 - \theta)\frac{1}{2}\|v\|^2$$

by the parallelogram law / Young's inequality.

Step 2: Write $\phi(x) = f(Ax - b)$ with A playing the role of the matrix and $-b$ the translation. This is exactly an affine composition $f(Ax + b)$ with the matrix A and constant vector $-b$.

By the result just proved, $\phi(x) = f(Ax + (-b))$ is convex. ✓

Explicit gradient and Hessian (bonus):

$$\nabla \phi(x) = A^\top (Ax - b), \quad \nabla^2 \phi(x) = A^\top A \succeq 0.$$

The Hessian $A^\top A$ is always positive semidefinite (it equals $\|Av\|^2/\|v\|^2 \geq 0$ along any direction v), confirming convexity. If A has full column rank, $A^\top A \succ 0$ and ϕ is *strictly* convex with a unique minimizer $x^* = (A^\top A)^{-1}A^\top b$ (the least-squares solution).

Q16 (Long Answer) — Time-Frequency Uncertainty

Explain the time-frequency trade-off (uncertainty principle) in signal processing terms. Then analyze the discrete-time signal $x = [1, -1, 1, -1, 2, 2, 2, 2]$ ($N = 8$).

- Identify (qualitatively) a “high-frequency” component and a “DC/low-frequency” component in x .
- Explain why a single global FFT summary can obscure *when* each component occurs.

Solution: Time-frequency uncertainty principle.

The *uncertainty principle* (Heisenberg-Gabor) states that a signal cannot be simultaneously well-localized in both time and frequency:

$$\Delta t \cdot \Delta \omega \geq \frac{1}{2},$$

where Δt is the time spread and $\Delta \omega$ is the frequency spread of the signal. In discrete signal processing terms:

- A signal concentrated in a *short time window* (like a spike) has a *broad frequency spectrum*.
- A signal with a *narrow frequency band* (like a pure sinusoid) must extend over a *long time interval*.

There is an inherent trade-off: you cannot simultaneously resolve both “when” and “at what frequency” without some blurring in one domain.

Analysis of $x = [1, -1, 1, -1, 2, 2, 2, 2]$, $N = 8$.

Observe the structure by splitting x :

- **Samples 1–4:** $[1, -1, 1, -1]$ — alternating signs at every sample, which is a *high-frequency* (Nyquist-rate) oscillation at frequency $k = N/2 = 4$ (i.e., $\omega = \pi$, the maximum discrete frequency).
 - **Samples 5–8:** $[2, 2, 2, 2]$ — a constant block, which is a pure *DC* (*zero-frequency*) or very low-frequency component.
- (a) **High-frequency component:** The alternating $[1, -1, 1, -1]$ in samples 1–4 is a *discrete cosine/sine wave at frequency* $k = 4$ (half the sampling rate). Each successive sample flips sign — this is the highest possible frequency in an 8-point signal.

DC/Low-frequency component: The constant $[2, 2, 2, 2]$ in samples 5–8 contributes purely to the $k = 0$ (DC) bin. It has no oscillation whatsoever.

- (b) **Why does a single global FFT obscure timing?**

The N -point Discrete Fourier Transform (DFT) computes:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

Each DFT coefficient $X[k]$ is a *global* sum over all time samples $n = 0, \dots, N-1$. It aggregates contributions from the entire signal, with no information about *when* each frequency component was present.

In our signal: the DFT will show energy at $k = 0$ (DC, from samples 5–8) and at $k = 4$ (high freq, from samples 1–4), but the spectrum *cannot tell you* that the high-frequency oscillation came first (samples 1–4) and the DC came second (samples 5–8). Both contributions are blended into the global coefficients.

Solution — Short-Time Fourier Transform (STFT): Apply the DFT to short overlapping windows of the signal. This sacrifices some frequency resolution (due to the uncertainty principle) but gains temporal localization: you can see which frequency appears in which time window.

Example DFT coefficients (approximate):

For $x = [1, -1, 1, -1, 2, 2, 2, 2]$:

$$X[0] = \sum_{n=0}^7 x[n] = 1 - 1 + 1 - 1 + 2 + 2 + 2 + 2 = 8 \quad (\text{DC total sum}).$$

$$X[4] = \sum_{n=0}^7 x[n](-1)^n = 1(1) + (-1)(-1) + 1(1) + (-1)(-1) + 2(1) + 2(-1) + 2(1) + 2(-1) = 1 + 1 + 1 + 1 + 2 - 2 + 2 - 2 = 4$$

The DFT shows nonzero at $k = 0$ and $k = 4$, but nothing about which time indices caused which.

Q17 (Long Answer) — Fenchel Conjugates and Convexification

- (a) Compute the Fenchel conjugate of $f(x) = x^2$, i.e.,

$$f^*(y) = \sup_{x \in \mathbb{R}} (yx - f(x)).$$

- (b) Compute the biconjugate f^{**} and show that $f^{**} = f$ for this convex f .
(c) Give a qualitative explanation of why, for a *nonconvex* function (e.g., a double-well shape), f^{**} typically equals the *convex envelope* of f rather than f itself.

Solution:

- (a) **Computing $f^*(y)$ for $f(x) = x^2$.**

$$f^*(y) = \sup_{x \in \mathbb{R}} [yx - x^2].$$

The function $\phi(x) = yx - x^2$ is concave in x (the coefficient of x^2 is $-1 < 0$), so its supremum is attained at the unique stationary point. Taking the derivative and setting to zero:

$$\frac{d}{dx} [yx - x^2] = y - 2x = 0 \implies x^* = \frac{y}{2}.$$

Substituting back:

$$f^*(y) = y \cdot \frac{y}{2} - \left(\frac{y}{2}\right)^2 = \frac{y^2}{2} - \frac{y^2}{4} = \frac{y^2}{4}.$$

$$\boxed{f^*(y) = \frac{y^2}{4}.$$

- (b) **Computing the biconjugate f^{**} .**

The biconjugate is the conjugate of the conjugate:

$$f^{**}(x) = (f^*)^*(x) = \sup_{y \in \mathbb{R}} [xy - f^*(y)] = \sup_{y \in \mathbb{R}} \left[xy - \frac{y^2}{4} \right].$$

Again, this is concave in y , so:

$$\frac{d}{dy} \left[xy - \frac{y^2}{4} \right] = x - \frac{y}{2} = 0 \implies y^* = 2x.$$

Substituting:

$$f^{**}(x) = x(2x) - \frac{(2x)^2}{4} = 2x^2 - x^2 = x^2.$$

$$\boxed{f^{**}(x) = x^2 = f(x).$$

This confirms the Fenchel–Moreau theorem: for a proper, closed, convex function, $f^{**} = f$.

(c) **Biconjugate as convex envelope for nonconvex f .**

For a nonconvex function, the Fenchel–Moreau theorem no longer guarantees $f^{**} = f$. Instead:

General fact: For *any* function f (convex or not), $f^{**}(x) \leq f(x)$ everywhere, and f^{**} is always convex and lower semi-continuous (as a pointwise supremum of affine functions). Moreover, f^{**} is the *largest* convex lower semi-continuous function that is $\leq f$ everywhere — this is called the *convex envelope* (or convex hull of the epigraph) of f .

Double-well example: Let $f(x) = (x^2 - 1)^2$ (two wells at $x = \pm 1$, a bump at $x = 0$). This function is not convex. The biconjugate $f^{**}(x)$ equals:

- 0 for $|x| \leq 1$ (the “filled-in” flat region connecting the two wells),
- $(x^2 - 1)^2$ for $|x| > 1$ (matches f outside the nonconvex region).

Geometrically, f^{**} is obtained by taking the *convex hull of the epigraph of f* — equivalently, “tightening a rubber band” around the underside of the graph. The nonconvex “dip” between the two wells is replaced by the horizontal line $y = 0$ connecting them.

Why this happens: The conjugate f^* captures only the “convex part” of f — it effectively discards the nonconvex portions by taking affine underestimators. When you take the conjugate twice, you get the tightest convex function lying below f , which is the convex envelope.

Practical implication: In optimization, solving $\min f$ where f is nonconvex is in general NP-hard. Replacing f by f^{**} (its convex envelope) gives a convex relaxation whose optimal value is a lower bound on the original. If the relaxation’s minimizer happens to satisfy $f^{**}(x^*) = f(x^*)$, it is also optimal for the original.

Q18 (Long Answer) — Dual Function and Weak Duality

Consider the equality-constrained problem:

$$\min_{x \in \mathbb{R}^n} x^\top x \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{p \times n}$ and $b \in \mathbb{R}^p$.

- Form the Lagrangian and derive the dual function $g(\nu)$ by minimizing over x .
- Write the dual optimization problem.
- Show (in words or inequalities) why $g(\nu)$ gives a *lower bound* on the primal optimal value p^* for every ν (weak duality).

Solution:

- (a) **Lagrangian and dual function.**

The problem has only equality constraints $Ax = b$ (no inequalities), so we introduce dual variables $\nu \in \mathbb{R}^p$:

$$L(x, \nu) = x^\top x + \nu^\top (Ax - b) = \|x\|_2^2 + \nu^\top Ax - \nu^\top b.$$

Minimize over x : $L(x, \nu)$ is a strictly convex quadratic in x (since the coefficient of $\|x\|^2$ is $1 > 0$). Taking the gradient and setting to zero:

$$\nabla_x L(x, \nu) = 2x + A^\top \nu = 0 \implies x^* = -\frac{1}{2} A^\top \nu.$$

Substitute $x^* = -\frac{1}{2} A^\top \nu$ back into L :

$$\begin{aligned} g(\nu) &= L(x^*, \nu) = \|x^*\|^2 + \nu^\top A x^* - \nu^\top b \\ &= \left\| -\frac{1}{2} A^\top \nu \right\|^2 + \nu^\top A \left(-\frac{1}{2} A^\top \nu \right) - \nu^\top b \\ &= \frac{1}{4} \nu^\top A A^\top \nu - \frac{1}{2} \nu^\top A A^\top \nu - \nu^\top b \\ &= -\frac{1}{4} \nu^\top A A^\top \nu - \nu^\top b. \end{aligned}$$

$$g(\nu) = -\frac{1}{4} \nu^\top A A^\top \nu - \nu^\top b.$$

This is a concave quadratic in ν (as expected: dual functions are always concave).

(b) Dual optimization problem.

Since there are no inequality constraints, there are no λ variables. The dual problem is:

$$\max_{\nu \in \mathbb{R}^p} g(\nu) = \max_{\nu \in \mathbb{R}^p} \left[-\frac{1}{4} \nu^\top A A^\top \nu - \nu^\top b \right].$$

This is a *concave* maximization problem (equivalently, minimize the convex $\frac{1}{4} \nu^\top A A^\top \nu + \nu^\top b$). The dual optimal value $d^* = \max_{\nu} g(\nu)$ satisfies $d^* \leq p^*$ in general (weak duality).

Stationarity of dual: Setting $\nabla_{\nu} g(\nu) = -\frac{1}{2} A A^\top \nu - b = 0$ gives the dual optimal $\nu^* = -2(A A^\top)^{-1} b$ (when $A A^\top$ is invertible). One can verify that this problem satisfies strong duality: $d^* = p^*$.

(c) Weak duality: $g(\nu) \leq p^*$ for all ν .

Argument:

For any dual variable $\nu \in \mathbb{R}^p$ and any *feasible* primal point $\tilde{x}$ (satisfying $A\tilde{x} = b$):

$$\begin{aligned} g(\nu) &= \inf_x L(x, \nu) \\ &\leq L(\tilde{x}, \nu) \quad (\text{infimum is a lower bound for any particular } x) \\ &= \tilde{x}^\top \tilde{x} + \nu^\top (A\tilde{x} - b) \\ &= \tilde{x}^\top \tilde{x} + \nu^\top \cdot 0 \quad (\text{since } A\tilde{x} = b) \\ &= f_0(\tilde{x}). \end{aligned}$$

So $g(\nu) \leq f_0(\tilde{x})$ for *every* dual ν and every feasible primal $\tilde{x}$.

Taking the infimum over all feasible $\tilde{x}$:

$$g(\nu) \leq \inf_{\tilde{x}: A\tilde{x}=b} f_0(\tilde{x}) = p^*.$$

Therefore $g(\nu) \leq p^*$ for all $\nu \in \mathbb{R}^p$. This is **weak duality**.

Interpretation: Every dual variable ν gives a lower bound on the true optimum p^* . Maximizing over ν gives the best (tightest) lower bound $d^* = \max_{\nu} g(\nu) \leq p^*$. When $d^* = p^*$, we say *strong duality* holds (which is the case here, since the problem is convex and Slater's condition is satisfied: the constraint is affine).

Concretely in our problem: If A has full row rank and $b \in \text{range}(A)$:

$$p^* = b^\top (AA^\top)^{-1}b \quad (\text{the minimum norm solution: } x^* = A^\top (AA^\top)^{-1}b),$$

and $d^* = p^*$ (strong duality holds for convex programs with linear constraints).